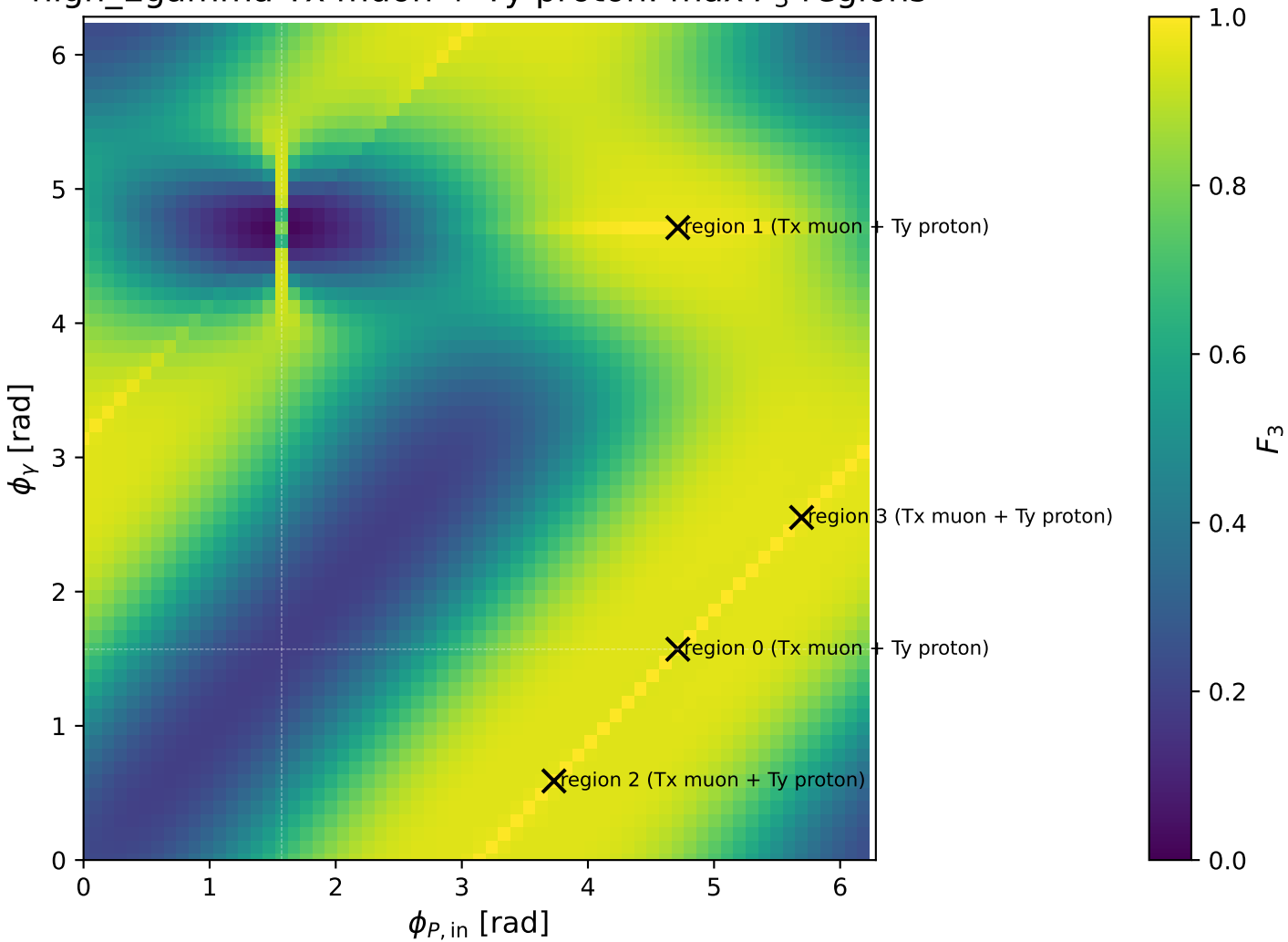
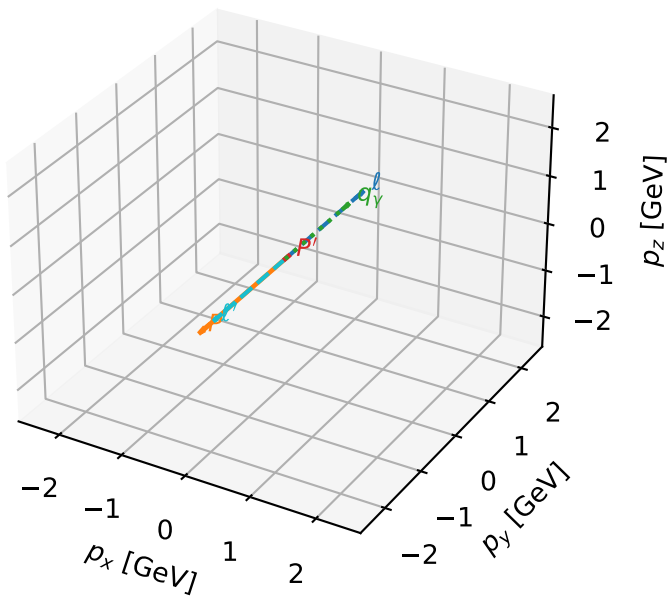


high\_Egamma Tx muon + Ty proton: max  $F_3$  regions



# Tx muon + Ty proton characteristic max $F_3$ configuration

3D momenta

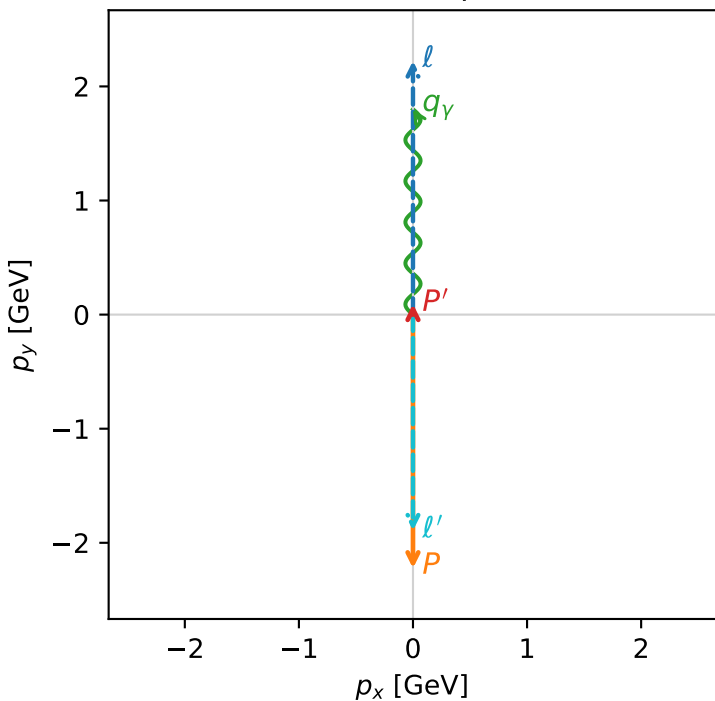


f3\_high\_egamma\_txtty\_region\_0 F\_3=0.999999  
 region: high\_Egamma\_TxTy\_region\_0  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=0.0929745$   
 $\theta_{in}=1.57079$ ,  $\phi_l=1.5708$ ,  $\phi_p=4.71239$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=1.5708$   
 $Q^2=16.8335$ ,  $x_B=0.846238$ ,  $t=-3.20247$ ,  $W^2=3.93848$ ,  $y=0.964473$   
 $\theta(l',\gamma)=180$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= 0.0000$   $p_y= 2.2231$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= -0.0000$   $p_y= -2.2231$   $p_z= 0.0000$   
 $l'$   $E= 1.8959$   $p_x= -0.0000$   $p_y= -1.8930$   $p_z= 0.0000$   
 $P'$   $E= 0.9426$   $p_x= 0.0000$   $p_y= 0.0930$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= 0.0000$   $p_y= 1.8000$   $p_z= 0.0000$

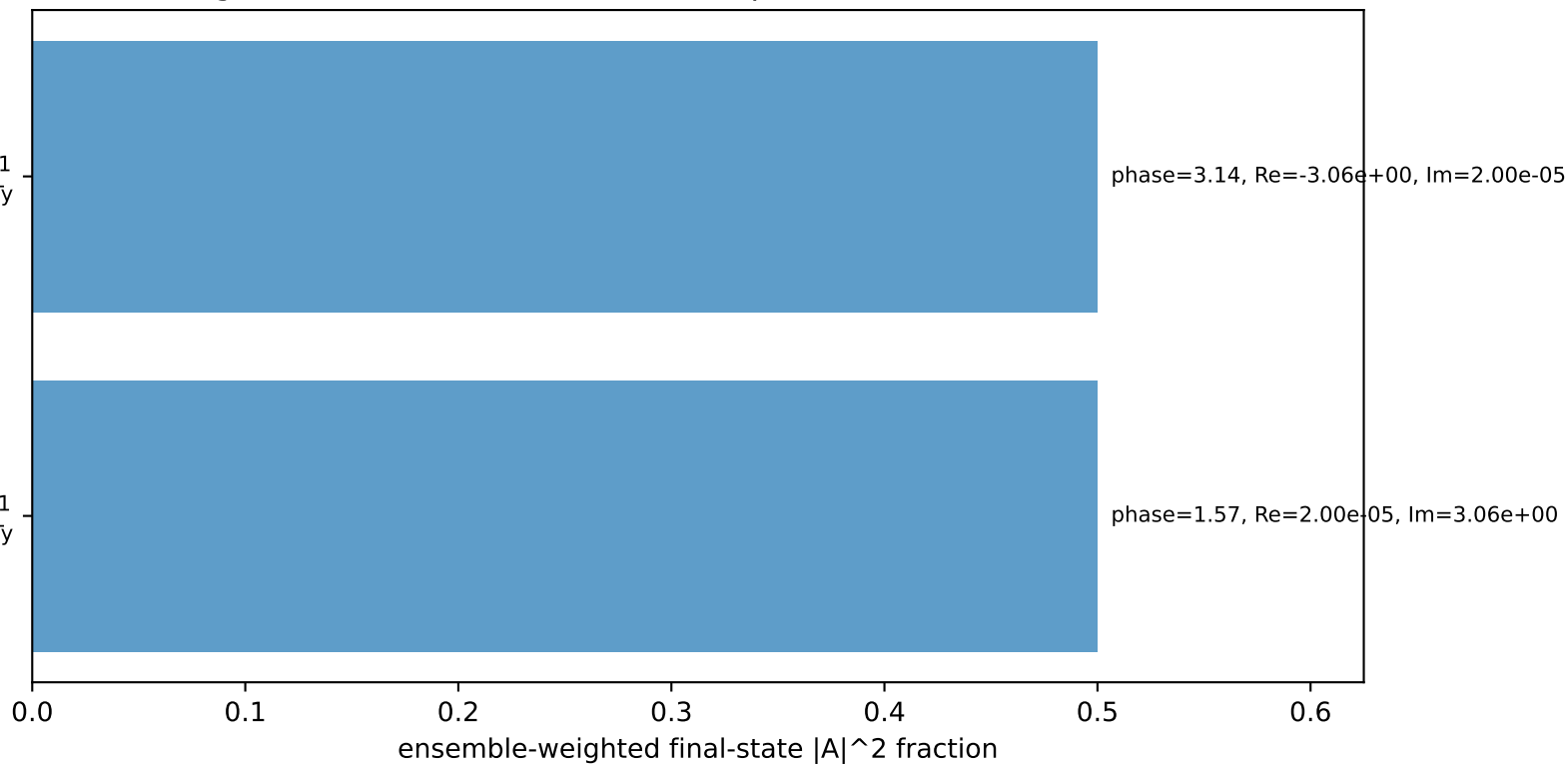
Transverse plane



$h_e=-1$ ,  $h_p=-1$ ,  $h_\gamma=+1$   
 electron Tx, proton Ty

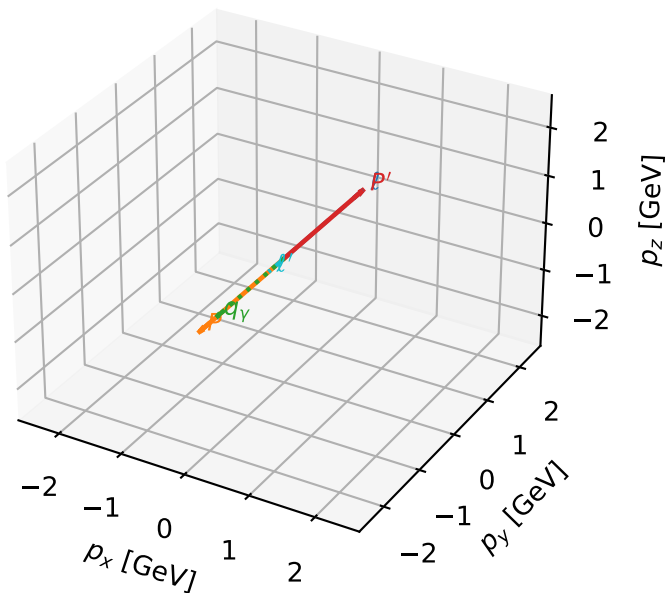
$h_e=+1$ ,  $h_p=+1$ ,  $h_\gamma=-1$   
 electron Tx, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



# Tx muon + Ty proton characteristic max $F_3$ configuration

3D momenta

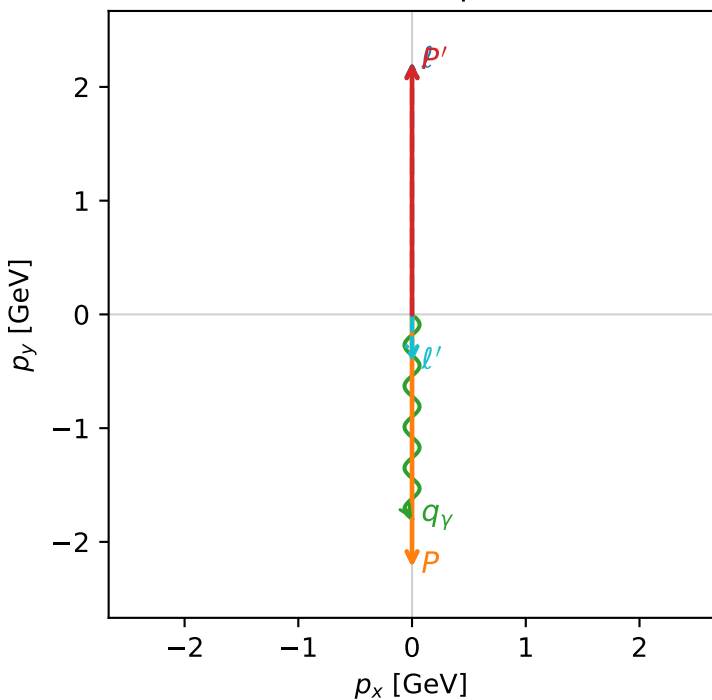


f3\_high\_egamma\_txtty\_region\_1 F\_3=0.999988  
 region: high\_Egamma\_TxTy\_region\_1  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=2.21757$   
 $\theta_{in}=1.57079$ ,  $\phi_l=1.5708$ ,  $\phi_p=4.71239$   
 $E_V=1.8$ ,  $\phi_V=4.71239$   
 $Q^2=3.75155$ ,  $x_B=0.183874$ ,  $t=-19.7196$ ,  $W^2=17.5311$ ,  $y=0.989236$   
 $\theta(l',\gamma)=0$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= 0.0000$   $p_y= 2.2231$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= -0.0000$   $p_y= -2.2231$   $p_z= 0.0000$   
 $l'$   $E= 0.4307$   $p_x= 0.0000$   $p_y= -0.4176$   $p_z= 0.0000$   
 $P'$   $E= 2.4078$   $p_x= 0.0000$   $p_y= 2.2176$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= -0.0000$   $p_y= -1.8000$   $p_z= 0.0000$

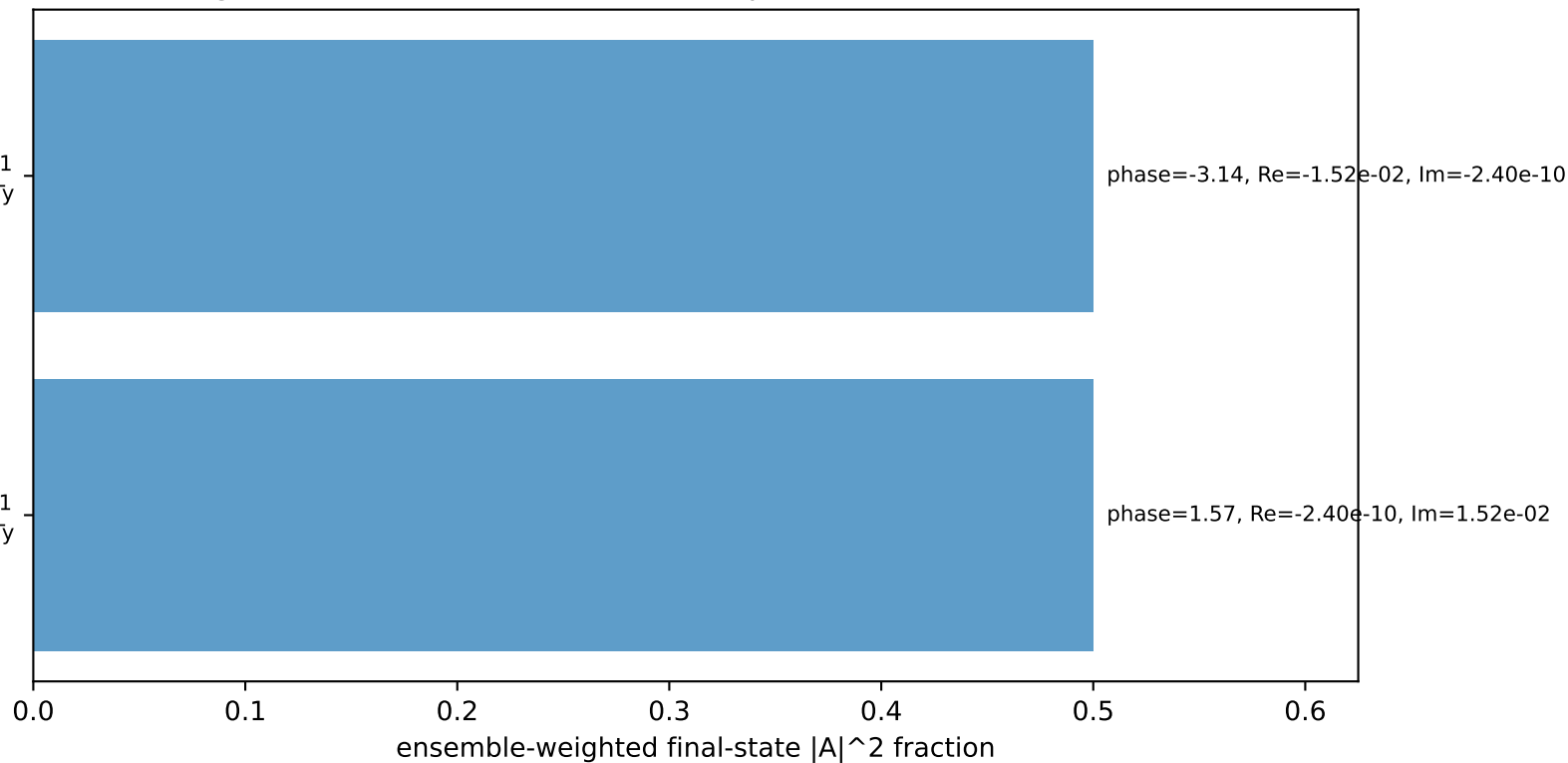
Transverse plane



$h_e=+1$ ,  $h_p=-1$ ,  $h_\gamma=-1$   
 electron Tx, proton Ty

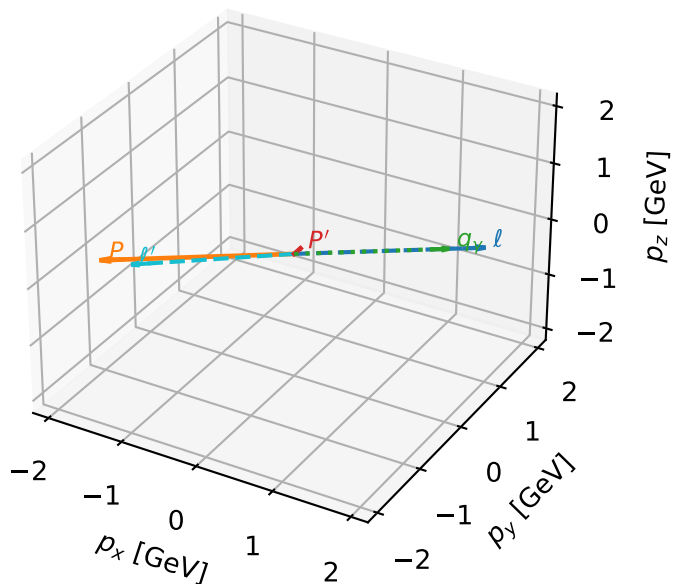
$h_e=-1$ ,  $h_p=+1$ ,  $h_\gamma=+1$   
 electron Tx, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



# Tx muon + Ty proton characteristic max $F_3$ configuration

3D momenta

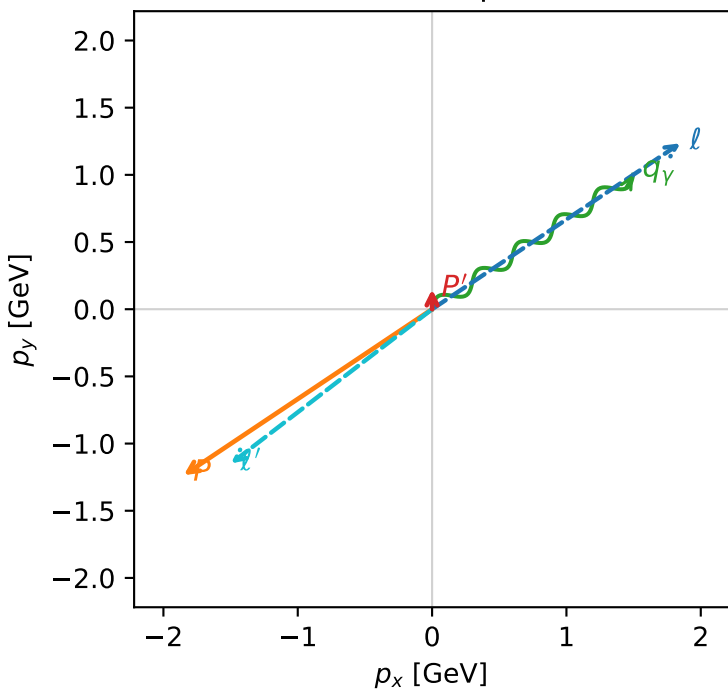


f3\_high\_egamma\_txtty\_region\_2 F\_3=0.999649  
 region: high\_Egamma\_TxTy\_region\_2  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

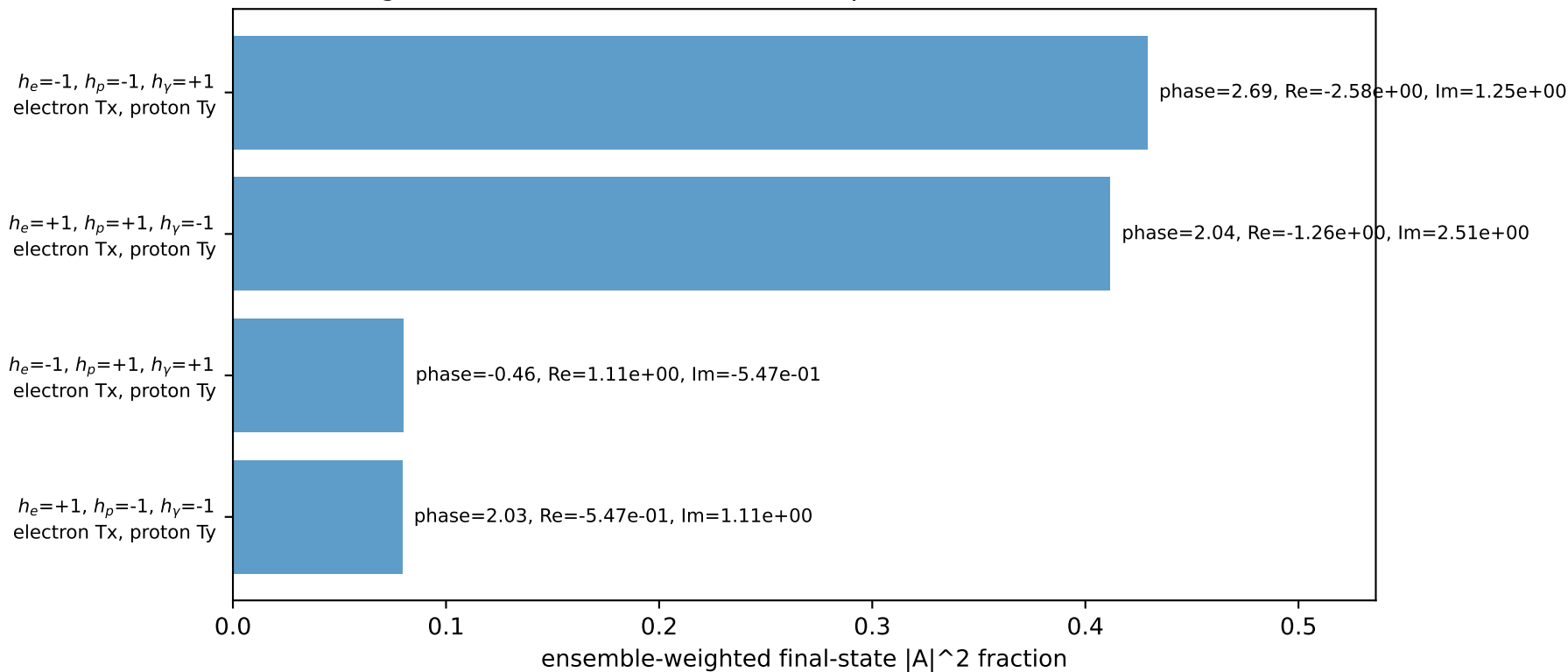
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=0.147625$   
 $\theta_{in}=1.57079$ ,  $\phi_l=0.589049$ ,  $\phi_p=3.73064$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=0.589049$   
 $Q^2=16.7538$ ,  $x_B=0.842878$ ,  $t=-3.18728$ ,  $W^2=4.00295$ ,  $y=0.963737$   
 $\theta(l', \gamma)=176.268$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= 1.8484$   $p_y= 1.2351$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= -1.8484$   $p_y= -1.2351$   $p_z= 0.0000$   
 $l'$   $E= 1.8890$   $p_x= -1.4966$   $p_y= -1.1477$   $p_z= 0.0000$   
 $P'$   $E= 0.9495$   $p_x= 0.0000$   $p_y= 0.1476$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= 1.4966$   $p_y= 1.0000$   $p_z= 0.0000$

Transverse plane

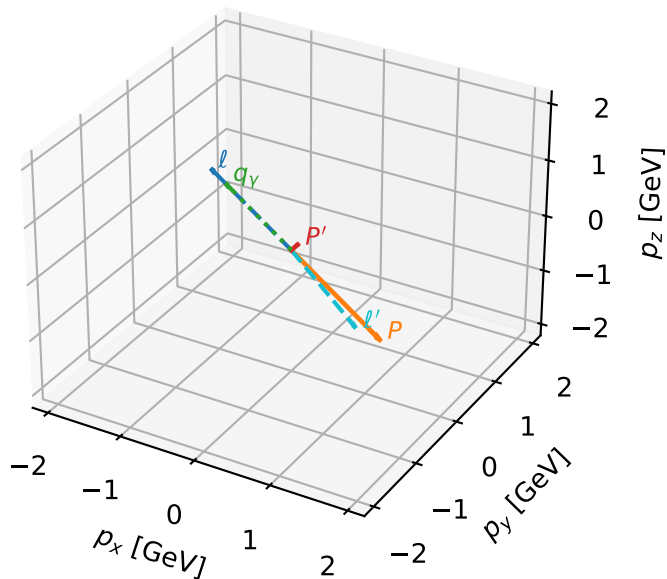


Leading initial-ensemble/final-state components (N=4, retained=100.0%)



# Tx muon + Ty proton characteristic max $F_3$ configuration

3D momenta



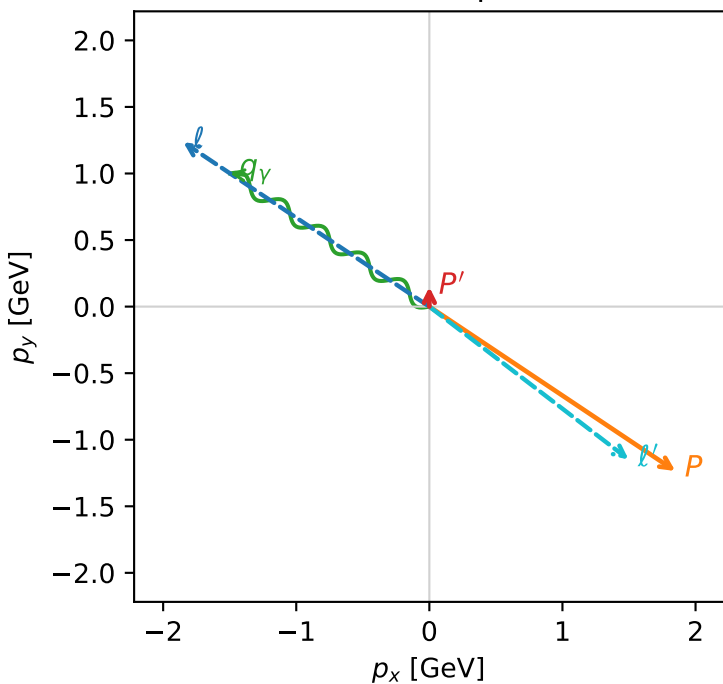
f3\_high\_egamma\_ttxty\_region\_3  $F_3=0.999568$   
 region: high\_Egamma\_TxTy\_region\_3  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=0.147625$   
 $\theta_{in}=1.57079$ ,  $\phi_l=2.55254$ ,  $\phi_p=5.69414$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=2.55254$   
 $Q^2=16.7538$ ,  $x_B=0.842878$ ,  $t=-3.18728$ ,  $W^2=4.00295$ ,  $y=0.963737$   
 $\theta(l',\gamma)=176.268$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

|            |             |                |                |                |
|------------|-------------|----------------|----------------|----------------|
| $l$        | $E= 2.2256$ | $p_x= -1.8484$ | $p_y= 1.2351$  | $p_z= -0.0000$ |
| $P$        | $E= 2.4129$ | $p_x= 1.8484$  | $p_y= -1.2351$ | $p_z= 0.0000$  |
| $l'$       | $E= 1.8890$ | $p_x= 1.4966$  | $p_y= -1.1477$ | $p_z= 0.0000$  |
| $P'$       | $E= 0.9495$ | $p_x= 0.0000$  | $p_y= 0.1476$  | $p_z= 0.0000$  |
| $q_\gamma$ | $E= 1.8000$ | $p_x= -1.4966$ | $p_y= 1.0000$  | $p_z= 0.0000$  |

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=100.0%)

